

CSC111 Final Exam Cheatsheet

Decimal, Binary and Hex

Decimal	Binary	Hex
0	0000	0x0
1	0001	0x1
2	0010	0x2
3	0011	0x3
4	0100	0x4
5	0101	0x5
6	0110	0x6
7	0111	0x7
8	1000	0x8
9	1001	0x9
10	1010	0xA
11	1011	0xB
12	1100	0xC
13	1101	0xD
14	1110	0xE
15	1111	0xF

Condition Flags

Program Status Register (PSR)

N	Z	C	V	Q	ICI/IT	T	Reserved	GE	Reserved	ICI/IT	ISR number
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► Negative bit

- N = 1 if most significant bit of result is 1

► Zero bit

- Z = 1 if all bits of result are 0

► Carry bit

- For unsigned addition, C = 1 if carry takes place
- For unsigned subtraction, C = 0 (carry = not borrow) if borrow takes place
- For shift/rotation, C = last bit shifted out

► oVerflow bit

- V = 1 if adding 2 same-signed numbers produces a result with the opposite sign
 - Positive + Positive = Negative, or
 - Negative + negative = Positive
- Non-arithmetic operations does not touch V bit, such as MOV, AND, LSL, MUL

Negative ----- signed result is negative

Zero ----- result is 0

Carry ----- add op → overflow
sub op doesn't borrow
last bit shifted out when shifting

oVerflow --- add/sub op → signed overflow

Carry and Overflow Flags w/ Arithmetic Instructions

Carry flag C = 1 (Borrow flag = 0) upon an **unsigned** addition if the answer is wrong (true result $> 2^{n-1}$)

Carry flag C = 0 (Borrow flag = 1) upon an **unsigned** subtraction if the answer is wrong (true result < 0)

Overflow flag V = 1 upon a **signed** addition or subtraction if the answer is wrong (true result $> 2^{n-1}$ or true result $< -2^{n-1}$)

Overflow may occur when adding 2 operands with the same sign, or subtracting 2 operands with different signs; Overflow cannot occur when adding 2 operands with different signs or when subtracting 2 operands with the same sign.

	Unsigned Addition	Unsigned Subtraction	Signed Addition or Subtraction
Carry flag	true result $> 2^{n-1} \rightarrow$ Carry flag=1 Borrow flag=0 (Result incorrect)	true result $< 0 \rightarrow$ Carry flag=0 Borrow flag=1 (Result incorrect)	N/A
Overflow flag	N/A	N/A	true result $> 2^{n-1}$ or true result $< -2^{n-1}$ $\rightarrow$ Overflow flag=1 (Result incorrect)

► 5

Condition Codes

Suffix	Description	Flags tested
EQ	E qual	Z=1
NE	N ot E qual	Z=0
CS/HS	Unsigned H igher or S ame	C=1
CC/LO	Unsigned L ower	C=0
MI	M inus (Negative)	N=1
PL	P lus (Positive or Zero)	N=0
VS	o verflow S et	V=1
VC	o verflow C leared	V=0
HI	Unsigned H igher	C=1 & Z=0
LS	Unsigned L ower or S ame	C=0 or Z=1
GE	Signed G reater or E qual	N=V
LT	Signed L ess T han	N!=V
GT	Signed G reater T han	Z=0 & N=V
LE	Signed L ess than or E qual	Z=1 or N!=V
AL	A lways	

► Condition Codes:

- EQ/NE: Z=1 / Z=0 (Equal/Not Equal)
- LT/GE: N≠V / N=V (Signed Less Than/Greater Equal)
- GT/LE: Z=0 & N=V / Z=1 or N≠V (Signed Greater/Less Equal)
- LO/HS: C=0 / C=1 (Unsigned Lower/Higher Same)
- HI/LS: C=1 & Z=0 / C=0 or Z=1 (Unsigned Higher/Lower Same)

► Flag Setting Instructions:

- CMP: R1 - R2 (result discarded)
- TST: R1 & R2 (result discarded)
- TEQ: R1 $\oplus$ R2 (result discarded)

- CMN: R1 + R2 (result discarded)

Branch Instructions

	Instruction	Description	Flags tested
Unconditional Branch	B <i>Label</i>	Branch to label	
Conditional Branch	BEQ <i>Label</i>	Branch if E Qual	Z = 1
	BNE <i>Label</i>	Branch if N ot E qual	Z = 0
	BCS/BHS <i>Label</i>	Branch if unsigned H igher or S ame	C = 1
	BCC/BLO <i>Label</i>	Branch if unsigned L ower	C = 0
	BMI <i>Label</i>	Branch if M inus (Negative)	N = 1
	BPL <i>Label</i>	Branch if P lus (Positive or Zero)	N = 0
	BVS <i>Label</i>	Branch if o V erflow S et	V = 1
	BVC <i>Label</i>	Branch if o V erflow C lear	V = 0
	BHI <i>Label</i>	Branch if unsigned H igher	C = 1 & Z = 0
	BLS <i>Label</i>	Branch if unsigned L ower or S ame	C = 0 or Z = 1
	BGE <i>Label</i>	Branch if signed G reater or E qual	N = V
	BLT <i>Label</i>	Branch if signed L ess T han	N != V
	BGT <i>Label</i>	Branch if signed G reater T han	Z = 0 & N = V
	BLE <i>Label</i>	Branch if signed L ess than or E qual	Z = 1 or N = ! V